

Article

Statistically Indistinguishable Performance of Lightweight CNNs with Explainable AI for Robust Orchid Disease Classification

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Abstract

Dendrobium Sonia orchid cultivation constitutes a vital commercial industry in Thailand; however, production remains persistently threatened by fungal and bacterial diseases. This study proposes a robust automated framework for orchid disease classification under conditions characterized by high visual uncertainty. A comparative analysis was conducted across four Convolutional Neural Network (CNN) architectures: ResNet-50 and three lightweight counterparts—MobileNetV3-Large, EfficientNetV2-B0, and NASNet-Mobile. All models were optimized using transfer learning, Cosine Decay scheduling, and EarlyStopping on a real-world dataset acquired from commercial orchid farms in Thailand. Experimental results indicate that ResNet-50 attained the highest overall performance (Accuracy: 98.96%, Macro F1: 0.9894, AUC-ROC: 0.9996), while EfficientNetV2-B0 achieved comparable results among the lightweight architectures (Accuracy: 98.47%, Macro F1: 0.9846, AUC-ROC: 0.9985). Importantly, statistical evaluation using the Wilcoxon Signed-Rank Test across five independent trials revealed no statistically significant difference between ResNet-50 and all three lightweight models ($p > 0.05$). This confirms the practical viability of deploying compact architectures on mobile platforms within smart farming systems without sacrificing diagnostic accuracy. Moreover, integrating Grad-CAM++ enhances interpretability by producing visual explanations that align with expert pathological assessments. This transparency effectively mitigates decision-making ambiguity and strengthens farmer confidence in adopting AI-driven precision agriculture.

Keywords: orchid disease classification; convolutional neural network; deep learning; Grad-CAM++



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1. Introduction

Dendrobium Sonia orchid is an economically significant crop that plays a vital role in Thailand's agricultural revenue. In 2023, Thailand was recognized as the world's second largest orchid exporter, generating total ornamental export revenue of USD 132 million, with orchids accounting for over 70% of this value and cut orchid exports alone reaching nearly USD 70 million [1]. Nevertheless, the orchid industry continues to face threats from numerous fungal and bacterial diseases, including Black rot, Yellow leaf spot, Leaf spot, Anthracnose, Stem rot, and Flower rusty spot, all of which have been reported to

negatively impact export value [2]. A further challenge is that disease diagnosis at the farm level still relies primarily on visual observation and the personal experience of individual farmers, without access to clear and definitive diagnostic information, resulting in a high risk of incorrect chemical application, deterioration of produce quality, and unnecessary cost burdens for farmers [3].

The application of deep learning—particularly Convolutional Neural Networks (CNNs)—has been demonstrated to perform with high effectiveness in classifying diseases across various crops. Mohanty et al. [4] pioneered this direction by showing that CNNs trained on the PlantVillage dataset could achieve 99.35% accuracy. However, that seminal work exhibited three primary limitations: (1) the unexplored feasibility of mobile deployment, (2) the absence of in-depth interpretability to elucidate model decisions, and (3) the lack of statistical stability testing. Subsequent work by Barman et al. [5] reported 98.29% accuracy for citrus leaf disease classification on mobile platforms, while Ferentinos [6] achieved 99.53% accuracy across 58 distinct plant–disease class combinations. Nevertheless, both studies still lack systematic robustness testing through inferential statistical analysis—a substantial hurdle in verifying model reliability prior to practical field deployment.

To address computational constraints in mobile deployment, the MobileNet family progressed from MobileNetV1 [7] to the more advanced MobileNetV3 [8], which integrates hard-swish activation functions and squeeze-and-excitation mechanisms. Sandler et al. [9] introduced MobileNetV2 with inverted residual structures to significantly reduce computational overhead. Tan and Le [10] proposed EfficientNet, achieving a balance between model depth, width, and resolution through compound scaling, later advancing it to EfficientNetV2 [11] for faster training and improved parameter efficiency. Furthermore, Zoph et al. [12] presented NASNet, which employs Neural Architecture Search (NAS) to automatically discover optimal architectural configurations for specific computer vision tasks.

Regarding model interpretability, Selvaraju et al. [13] pioneered Grad-CAM as a fundamental technique for generating visual saliency maps to identify influential regions in model predictions. This was refined into Grad-CAM++ by Chattopadhyay et al. [14], offering superior localization accuracy. The efficacy of Grad-CAM++ in agricultural contexts was validated by Pal et al. [15], while Pei et al. [16] demonstrated that compact models can achieve an optimal balance between computational efficiency and interpretability a principle that serves as a foundational framework for the methodology proposed in this study.

Despite these advances, the application of deep learning to orchid disease detection still presents three significant research gaps: a scarcity of high-quality image datasets reflecting diverse real-world conditions, the inherently high visual uncertainty arising from morphologically similar disease presentations [12–14], and the prevalent reliance on single experimental runs that neglects stochastic variance in AI training. To directly address each of these gaps, this study proposes an integrated framework emphasizing both predictive performance and decision-making transparency, achieved through four principal contributions.

First, to address the dataset scarcity gap, a standardized multi-class dataset was constructed from six orchid disease categories collected across 10 commercial farms in two major production provinces, providing the environmental diversity necessary to ensure real-world robustness and generalizability. Second, to mitigate the challenge of visual uncertainty among morphologically similar diseases, Grad-CAM++ saliency maps were integrated and validated against expert pathological diagnoses, directly confirming that model decisions are driven by disease-specific lesion features rather than spurious background ar-

tifacts. Third, to overcome the limitation of single-run evaluations, a stability validation protocol was implemented comprising five repeated experimental trials subjected to Wilcoxon Signed-Rank Test analysis, providing rigorous statistical evidence of model reliability. Fourth, bridging the gap between diagnostic accuracy and deployment feasibility, a comparative evaluation of ResNet-50 and three lightweight architectures— MobileNetV3-Large, EfficientNetV2-B0, and NASNet-Mobile—was conducted to identify an optimal balance between accuracy and computational efficiency for mobile-edge deployment.

Collectively, this framework reduces diagnostic ambiguity and delivers a high-impact, trustworthy tool that fosters farmer confidence and prevents costly chemical misapplication in practical field-level deployment.

2. Materials and Methods

To attain the objective of developing an orchid disease diagnostic system characterized by high precision and statistical reliability, this study implemented a systematic research framework. The methodology encompasses real-world dataset curation, comparative benchmarking of convolutional neural network (CNN) architectures, model robustness validation, and interpretability analysis of the decision-making process. The specific procedural steps are detailed as follows:

2.1. Data Acquisition and Distribution Analysis

The primary dataset was collected from ten commercial Dendrobium Sonia orchid farms in Nakhon Pathom and Samut Sakhon provinces, Thailand. Data collection was carried out between May 2024 and February 2025 through a strategic partnership with the Thai Orchid Growers Cooperative and local orchid experts. All images were captured at a high resolution of 5184 × 3456 pixels under standardized, controlled lighting conditions to clearly represent lesion characteristics. The final dataset comprises 1832 images distributed across six disease classes: (1) Black rot or Phytophthora, (2) Yellow leaf spot, (3) Leaf spot, (4) Anthracnose, (5) Stem rot, and (6) Flower rusty spot, as summarized in Table 1. Due to the near-uniform distribution across all classes, the dataset is considered inherently balanced; therefore, no additional balancing techniques were required during the data preparation phase.

Table 1. Dataset Distribution per Class.

Class	Number of Images	Percentage (%)
Black rot or Phytophthora	332	18.1
Yellow leaf spot	300	16.4
Leaf spot	300	16.4
Anthracnose	300	16.4
Stem rot	300	16.4
Flower rusty spot	300	16.4
Total	1832	100.0

To preserve class distribution proportions during the experimental phase, an 80:20 Stratified Split methodology was employed, consistent with established practices in deep learning-based image classification research [17,18]. This resulted in a training set of 1466 images and a test set of 366 images. The stratified splitting approach ensures that the relative proportion of each class is maintained across both subsets, thereby reducing evaluation bias and enhancing the reliability of model performance metrics [19]. Furthermore, the assessment of the Imbalance Ratio (IR), which was calculated to be 1.11, confirmed that the data distribution remained within an acceptable threshold of equilibrium [20]. Consequently, the implementation of synthetic oversampling techniques or algorithmic

class-weighting was precluded, as such interventions are warranted only when a significant disparity in class frequencies is detected [21,22], thereby mitigating potential model bias while maintaining the integrity of the original data distribution.

2.2. Data Preprocessing and Augmentation

All input images were standardized to a resolution of 224×224 pixels and subjected to min-max normalization, scaling pixel intensities to a numerical range 0 to 1 [23]. To enhance model generalization and mitigate overfitting, real-time data augmentation was implemented via the TensorFlow Data Pipeline (TensorFlow v2.16.2, Python v3.10.14) during the training phase [24]. The augmentation suite comprised a series of stochastic transformations, including random rotations ($\pm 30^\circ$), horizontal and vertical flips, and zoom variations ($\pm 20\%$) [24,25]. These transformations were applied dynamically in an “on-the-fly” manner during each training epoch, allowing the model to learn from a synthetically expanded and diverse feature space without incurring additional physical storage overhead [25]. Crucially, data augmentation was strictly confined to the training pipeline. The test dataset was subjected only to resizing and normalization to ensure that the evaluation metrics accurately reflect the model’s inference performance in real-world deployment scenarios [24].

2.3. Comparative Architectures and Benchmarking Protocol

2.3.1. Model Selection and Architectural Characteristics

This study evaluates four distinct CNN architectures, each pre-trained on the ImageNet dataset:

1. ResNet-50 (~23 M parameters): Employed as the primary baseline, as it represents the most widely adopted CNN reference architecture in plant disease detection research and has been consistently used as a comparative benchmark in prior studies [4,6], thereby enabling meaningful and direct comparison with the existing body of literature. Although more recent architectures such as Vision Transformers (ViT) and ConvNeXt have demonstrated superior performance on large-scale datasets, they yield accuracy levels comparable to ResNet when applied to medium-scale data [26]. Furthermore, transformer-based models are prone to overfitting on domain-specific datasets of constrained scale without sufficient domain-specific pretraining [27]. Moreover, the deep residual learning mechanism of ResNet-50 enables the extraction of subtle morphological features and complex disease patterns that shallower architectures may fail to capture, as reflected by its consistently strong performance across diverse visual classification tasks [28]. ResNet-50 therefore serves as an appropriate upper-bound reference for comparison against lightweight architectures, which constitutes the primary objective of this study.
2. MobileNetV3-Large (~5.4 M parameters) [8]: A high-performance lightweight architecture optimized for mobile deployment, incorporating hard-swish activation functions and squeeze-and-excitation modules to maximize classification accuracy under resource-constrained conditions.
3. EfficientNetV2-B0 (~7 M parameters) [11]: An architecture designed to balance computational efficiency and classification accuracy through compound scaling of network depth, width, and input resolution, making it particularly suitable for deployment scenarios requiring both high precision and reduced inference latency.
4. NASNet-Mobile (~5.3 M parameters) [12]: A model derived through Neural Architecture Search (NAS) to optimize mobile-scale efficiency, providing an automated architecture design perspective that complements the manually engineered lightweight models evaluated in this comparative study.

2.3.2. Dual-Phase Transfer Learning and Multi-Seed Protocol

A dual-phase Transfer Learning strategy was adopted:

Phase I (Feature Extraction): The weights of the pre-trained backbones were frozen, and only the newly integrated classification heads were trained for 20 epochs with an initial learning rate of 1×10^{-3} , reduced adaptively to a minimum of 1×10^{-6} . A ReduceLROnPlateau scheduler (factor = 0.5, patience = 3) was employed alongside an Early Stopping mechanism to mitigate potential overfitting.

Phase II (Fine-tuning): The entire network was unfrozen for comprehensive fine-tuning over 50 epochs with an initial learning rate of 1×10^{-5} , decaying to a minimum of 1×10^{-7} . A Cosine Decay learning rate scheduler was employed to ensure smooth convergence toward the global optimum.

To ensure statistical robustness and address the inherent stochasticity of deep learning models, all experiments were replicated using five distinct random seeds (42, 123, 256, 512, and 999). Performance metrics are reported as Mean $\pm$ Standard Deviation (SD), adhering to rigorous AI evaluation standards.

2.3.3. Computational Infrastructure and Latency Benchmarking

The experimental framework was implemented on a platform featuring an Apple M3 processor with 24 GB of unified memory and an integrated Apple Metal GPU (Apple Inc., Cupertino, CA, USA). The software environment was configured with macOS Tahoe 26.4 (arm64), Python 3.10.14, and TensorFlow 2.16.2.

To evaluate edge-device deployment feasibility, all models were converted to INT8-quantized TensorFlow Lite (TFLite) format. Inference benchmarks were subsequently conducted on the Apple M3 CPU, serving as a high-performance proxy for mobile hardware latency. Benchmarking involved 100 independent inference cycles with a batch size of one to characterize real-time processing capabilities. The resulting computational metrics are synthesized in Table 2.

Table 2. Computational efficiency and deployment benchmarks of optimized models.

Model	TFLite Size (MB)	Training Time per Run (min)	CPU Latency (ms)	Throughput (FPS)
ResNet-50	23.14	11.8 ± 1.2	23.89 ± 0.08	41.9
MobileNetV3-Large	3.33	6.9 ± 0.5	7.15 ± 0.02	139.8
EfficientNetV2-B0	6.79	10.2 ± 1.4	23.21 ± 0.11	43.1
NASNet-Mobile	5.09	15.6 ± 1.5	27.81 ± 1.11	36.0

Training time is averaged across five seeds. CPU latency is reported as mean $\pm$ SD over 100 inference runs using INT8-quantized TFLite models (batch size = 1).

2.3.4. Performance Evaluation Metrics

The classification performance of each architecture was evaluated using four standard metrics derived from the confusion matrix. Let TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively. The metrics are formally defined as follows:

Accuracy measures the overall proportion of correctly classified instances across all disease categories:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

Precision measures the proportion of correct positive predictions among all positive predictions made by the model:

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

Recall measures the proportion of actual positive instances correctly identified by the model:

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

F1-Score is the harmonic mean of Precision and Recall, providing a balanced performance measure particularly suited to datasets exhibiting class imbalance:

$$F1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (4)$$

All metrics were computed using macro-averaging across all six orchid disease classes to ensure equal contribution from each class regardless of sample size.

2.4. Explainable AI (XAI) via Grad-CAM++ Analysis

To facilitate model interpretability, Grad-CAM++ was integrated to generate Class Activation Heatmaps from the final convolutional layer of each architecture. This technique provides empirical validation that the models' diagnostic decisions are derived from intrinsic pathological lesion features rather than spurious background artifacts. This XAI analysis serves as critical evidence base for transparency and trust before the deployment of the diagnostic system in actual agricultural environments.

To quantitatively assess the spatial concentration of model attention, the Activation Energy Ratio (AER) was proposed, inspired by the energy concentration framework introduced by Zhou et al. [29]. Given the absence of pixel-level lesion annotations in the dataset, foreground disease regions were adaptively defined by applying Otsu thresholding [30] to the normalized Grad-CAM++ heatmap, thereby serving as an annotation-free proxy for discriminative lesion regions. The AER is formally defined as follows:

$$AER = \frac{\sum_{(i,j) \in \Omega_{fg}} H(i,j)}{\sum_{(i,j) \in all} H(i,j)} \times 100\% \quad (5)$$

where $H(i, j)$ denotes the Grad-CAM++ activation intensity at spatial location (i, j) , and Ω_{fg} represents the foreground lesion region identified via Otsu adaptive thresholding. Localization performance was categorized into three levels based on AER values: Correct Localization ($AER \geq 70\%$), Partial Localization ($50\% \leq AER < 70\%$), and Incorrect Localization ($AER < 50\%$). An AER value of 70% or above indicates that the model's attention is predominantly concentrated on pathologically relevant disease structures, thereby providing objective evidence of localization precision.

2.5. Statistical Significance Testing

To determine whether the performance disparities between architectures were statistically significant, the Wilcoxon Signed-Rank Test was utilized for pairwise comparisons. ResNet-50 was designated as the reference baseline, representing the most computationally complex architecture in the study. This non-parametric approach is particularly suitable for small sample sizes ($n = 5$) as it does not require the assumption of a normal distribution. The significance level was set at $\alpha = 0.05$ to rigorously evaluate whether lightweight architectures can achieve performance levels statistically comparable to or exceeding those of larger-scale models.

3. Results

3.1. Comparative Analysis of CNN Architectures' Performance

The performance efficacy of four CNN architectures, including ResNet-50 and three computationally efficient models: MobileNetV3-Large, EfficientNetV2-B0, and NASNet-Mobile, is presented in Table 3. Performance metrics are reported as Mean $\pm$ Standard Deviation (SD) across five experimental iterations with stochastic seed initializations ($n = 5$ seeds) to ensure the statistical significance and robustness of the findings.

Table 3. Comparative performance of four CNN architectures (Mean $\pm$ SD, $n = 5$ seeds, 80/20 split).

Architecture	Accuracy	Macro F1	Macro Precision	Macro Recall	Weighted F1	AUC-ROC
ResNet-50	0.9896 $\pm$ 0.0097	0.9894 $\pm$ 0.0098	0.9896 $\pm$ 0.0098	0.9871 $\pm$ 0.0106	0.9919 $\pm$ 0.0092	0.9996 $\pm$ 0.0001
MobileNetV3-Large	0.9749 $\pm$ 0.0088	0.9743 $\pm$ 0.0106	0.9748 $\pm$ 0.0074	0.9754 $\pm$ 0.0181	0.9737 $\pm$ 0.0090	0.9994 $\pm$ 0.0003
EfficientNetV2-B0	0.9847 $\pm$ 0.0054	0.9846 $\pm$ 0.0060	0.9847 $\pm$ 0.0054	0.9843 $\pm$ 0.0063	0.9851 $\pm$ 0.0056	0.9985 $\pm$ 0.0009
NASNet-Mobile	0.9759 $\pm$ 0.0082	0.9759 $\pm$ 0.0080	0.9760 $\pm$ 0.0082	0.9763 $\pm$ 0.0083	0.9763 $\pm$ 0.0090	0.9990 $\pm$ 0.0004

The experimental results demonstrate that ResNet-50 outperformed other candidates across all evaluated metrics, achieving a peak accuracy of 0.9896 (SD = 0.0097) and a near-optimal AUC-ROC of 0.9996 (SD = 0.0001). These results underscore its superior classification capabilities and convergence stability. EfficientNetV2-B0 followed as the second-best model with an accuracy of 0.9847 (SD = 0.0054). Notably, this architecture exhibited the lowest variance among the cohort, signifying exceptional robustness and generalization stability against variations in initial seed values, a crucial attribute for dependable soft computing applications.

Furthermore, NASNet-Mobile and MobileNetV3-Large yielded comparable performance levels, with accuracies of 0.9759 (SD = 0.0082) and 0.9749 (SD = 0.0088), respectively. Although MobileNetV3-Large recorded the lowest accuracy in this study, its architectural efficiency and reduced computational overhead provide a strategic advantage for real-time deployment on resource-constrained edge devices and mobile platforms, as illustrated in Figure 1.

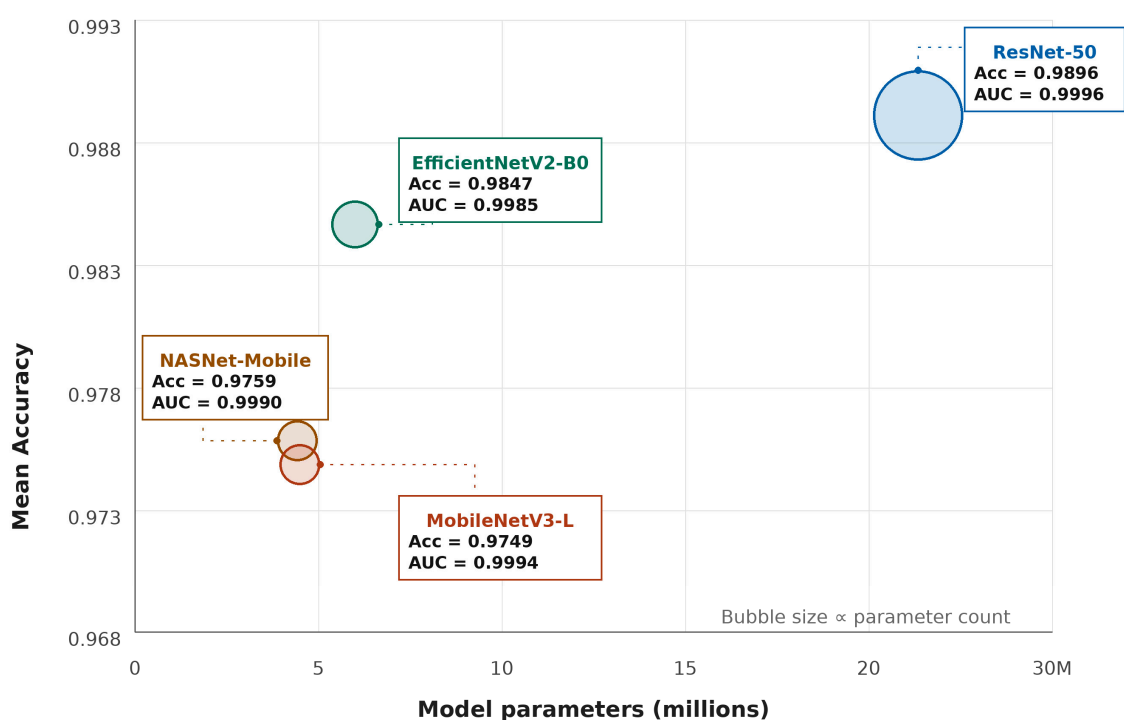


Figure 1. Accuracy vs. model size. Bubble diameter $\propto$ trainable parameters. AUC-ROC in parentheses.

3.2. Class-Wise Performance Comparison and Analysis

Table 4 details the class-wise F1-scores for the four evaluated CNN architectures. The experimental findings reveal that all models achieved perfect classification (F1-score = 1.0000) for Black rot, Stem rot, and Flower rusty spot across all iterations. This consistent performance indicates that these specific diseases possess highly discriminative morphological features, allowing the architectures to extract robust and unique visual descriptors for unambiguous classification.

Table 4. Class-wise F1-score (Mean $\pm$ SD, $n = 5$ seeds).

Architecture	Black Rot	Yellow Leaf Spot	Leaf Spot	Anthracnose	Stem Rot	Flower Rusty Spot
ResNet-50	1.0000 $\pm$ 0.0000	0.9509 $\pm$ 0.0247	0.9682 $\pm$ 0.0290	0.9825 $\pm$ 0.0075	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
MobileNetV3-Large	0.9909 $\pm$ 0.0095	0.9306 $\pm$ 0.0191	0.9448 $\pm$ 0.0163	0.9771 $\pm$ 0.0116	0.9981 $\pm$ 0.0019	1.0000 $\pm$ 0.0000
EfficientNetV2-B0	0.9911 $\pm$ 0.0089	0.9596 $\pm$ 0.0158	0.9643 $\pm$ 0.0162	0.9882 $\pm$ 0.0083	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
NASNet-Mobile	0.9943 $\pm$ 0.0071	0.9344 $\pm$ 0.0193	0.9316 $\pm$ 0.0188	0.9911 $\pm$ 0.0066	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000

Conversely, Yellow leaf spot and Leaf spot emerged as the most challenging categories for all architectures, primarily due to significant inter-class visual similarity and overlapping visual symptoms during the early stages of infection, as evidenced by the confusion matrix in Figure 2. ResNet-50 demonstrated a clear competitive advantage in handling these complex classes, attaining F1-scores of 0.9509 (SD = 0.0247) for Yellow leaf spot and 0.9682 (SD = 0.0290) for Leaf spot. In contrast, NASNet-Mobile exhibited the lowest discriminative power for these two categories, with F1-scores of 0.9344 (SD = 0.0193) and 0.9316 (SD = 0.0188), respectively.

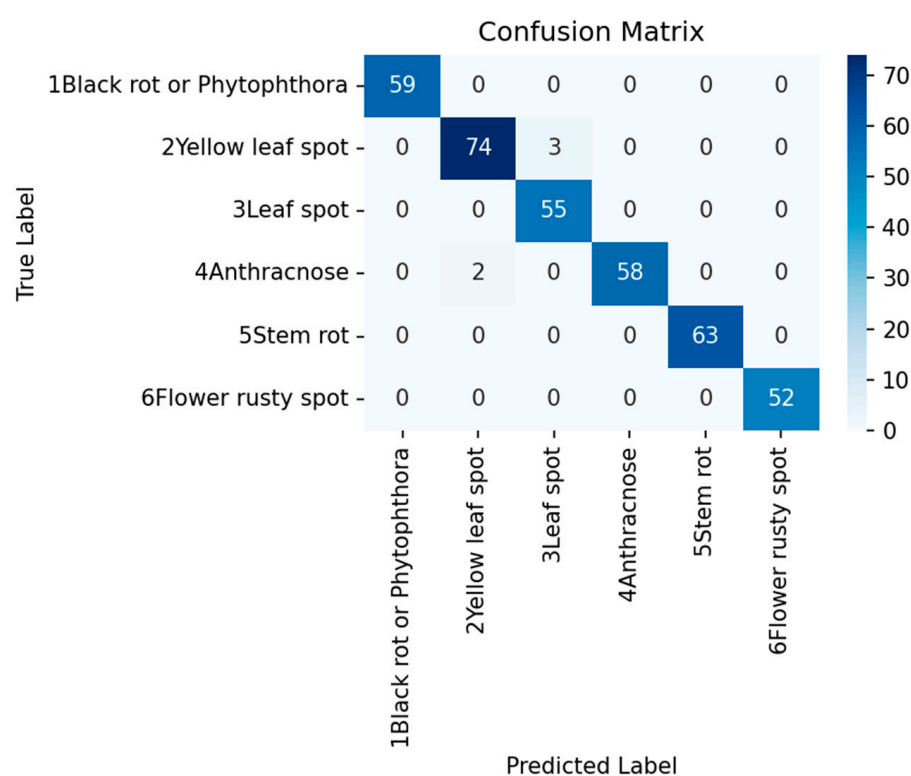


Figure 2. Confusion matrix of ResNet-50.

These results suggest that the performance variance among the studied architectures is predominantly driven by their capacity for fine-grained classification. While all models excel at identifying diseases with distinctive symptoms, deeper architectures like ResNet-50

prove more effective at capturing subtle, high-level features necessary to differentiate between morphologically similar disease patterns compared to their lightweight counterparts.

3.3. Statistical Significance Analysis Using Wilcoxon Signed-Rank Test

To rigorously evaluate the performance variances among the candidate architectures, a non-parametric Wilcoxon Signed-Rank Test was conducted, with ResNet-50 serving as the baseline for all comparative pairs. As summarized in Table 5, the statistical analysis revealed that the performance differences between ResNet-50 and all other architectures were not statistically significant ($p > 0.05$).

Table 5. Wilcoxon Signed-Rank Test results (Accuracy, $n = 5$, $\alpha = 0.05$).

Comparison	W-Statistic	p-Value	Result
ResNet-50 vs MobileNetV3-Large	W = 6.0	0.1250	Not significant ($p \geq 0.05$)
ResNet-50 vs EfficientNetV2-B0	W = 5.0	0.5000	Not significant ($p \geq 0.05$)
ResNet-50 vs NASNet-Mobile	W = 5.5	0.0625	Not significant ($p \geq 0.05$) *

* Note: The comparison between ResNet-50 and NASNet-Mobile yielded a borderline p -value of 0.0625, which approaches but does not reach statistical significance at $\alpha = 0.05$. Given the limited number of random seeds ($n = 5$), this result should be interpreted with caution, as a larger seed set may affect the statistical conclusion for this specific comparison [31,32].

Specifically, the comparison between ResNet-50 and EfficientNetV2-B0 yielded the highest p -value of 0.5000, indicating the highest degree of performance parity between the two models. Conversely, the pair comprising ResNet-50 and NASNet-Mobile exhibited the lowest p -value of 0.0625; while this remains above the predefined significance threshold, it is notably borderline and approaches statistical significance, necessitating a cautious interpretation of this specific comparison. These findings carry profound practical implications, demonstrating that lightweight architectures, namely MobileNetV3-Large, EfficientNetV2-B0, and NASNet-Mobile, with model sizes ranging from only 17–29 MB, can achieve performance levels statistically indistinguishable from the 98 MB ResNet-50 without significant loss in classification accuracy.

3.4. Interpretability and Visual Explanation Analysis via Grad-CAM++

To ensure model transparency and facilitate “Explainable AI” (XAI), Grad-CAM++ was employed to generate class activation maps for visual validation. The quantitative analysis for MobileNetV3-Large confirmed “Correct Localization” efficacy, with over 70% of the activation energy concentrated on the actual lesion areas across five of the six classes: Black rot, Leaf spot, Anthracnose, Stem rot, and Flower rusty spot, as illustrated in Figures 3–6. These results demonstrate the model’s high precision in delineating pathological regions.

For the Yellow leaf spot class, “Partial localization” was observed, characterized by activation maps spreading across the lesions and their immediate vicinities (Figure 4). This observation aligns with the diffused morphological characteristics of the disease. Crucially, no “Incorrect localization” was detected in any experimental trials, validating that the architectures learned discriminative features intrinsically linked to disease morphology rather than background artifacts.

Deep visual analysis further elucidates the performance disparities among architectures, particularly under conditions characterized by high visual uncertainty. ResNet-50 exhibited highly concentrated activation maps on lesion boundaries and distinct chlorotic transitions (Figure 3). This suggests that its architectural depth (23 M parameters) enables the extraction of superior fine-grained features. In contrast, NASNet-Mobile displayed broader activation patterns (Figure 6), indicating that its specific “Cell” architecture might

be more attuned to global texture representations rather than localized, high-resolution feature differentiation.

Grad-CAM++ Visual Explanation — ResNet-50

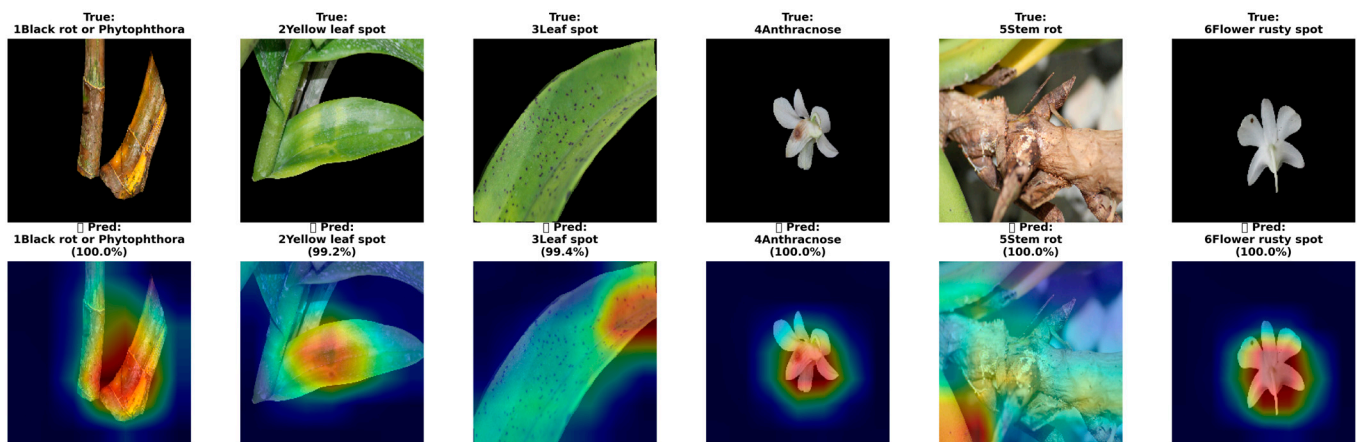


Figure 3. Grad-CAM++ visual explanation—ResNet-50.

Grad-CAM++ Visual Explanation — MobileNetV3-Large

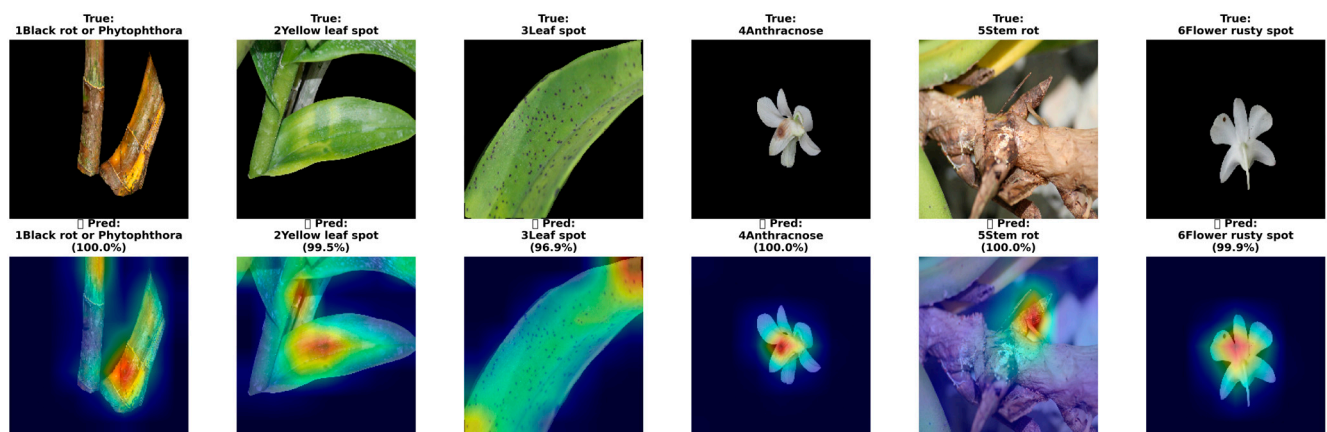


Figure 4. Grad-CAM++ visual explanation—MobileNetV3-Large.

Grad-CAM++ Visual Explanation — EfficientNetV2-B0

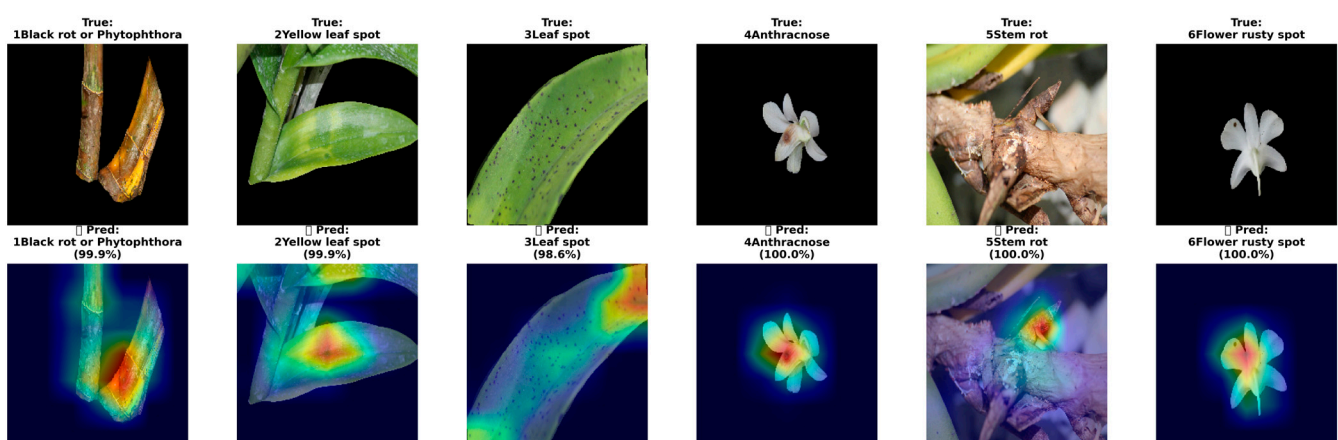


Figure 5. Grad-CAM++ visual explanation—EfficientNetV2-B0.

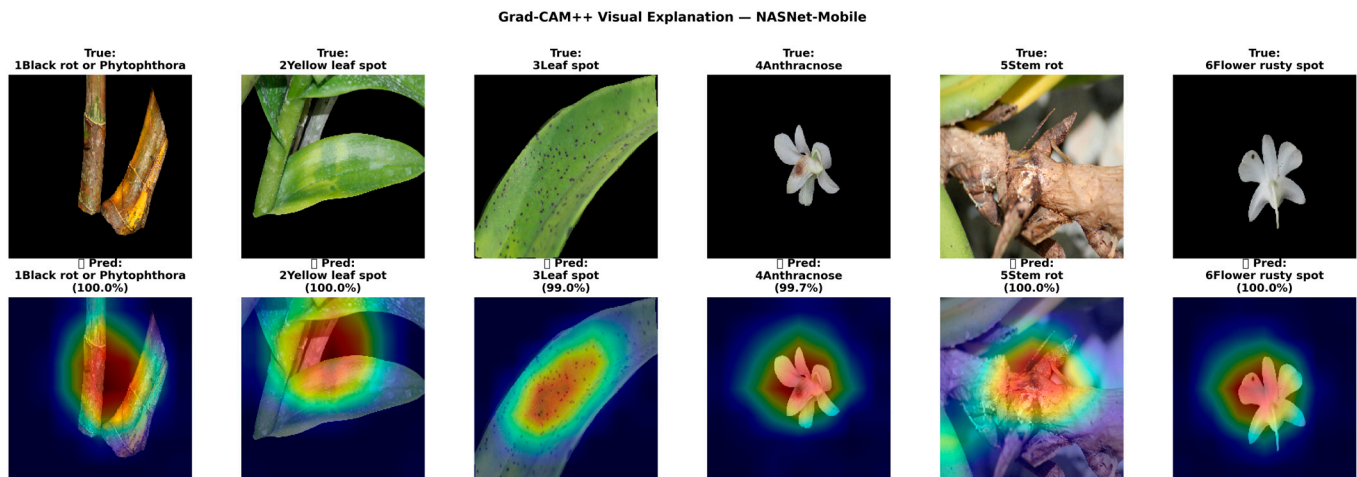


Figure 6. Grad-CAM++ visual explanation—NASNet-Mobile.

4. Discussion

The results of this study challenge the assumption that larger architectures always provide superior performance in the context of plant disease classification. Although ResNet-50 demonstrated the highest accuracy (0.9896 ± 0.0097), the lack of statistical significance compared to all compact architectures (Wilcoxon Signed-Rank Test, $p > 0.05$) suggests that model selection for field deployment should not be based solely on accuracy metrics. Instead, emphasis should be placed on the balance between performance, size, and deployability. This aligns with the concept proposed by Zhang et al. [33], which emphasizes that selection criteria for practical applications must encompass deployment readiness.

This study presents an integrated comparative framework for orchid disease classification that combines five-fold statistical validation, Wilcoxon Signed-Rank significance testing, Grad-CAM++ XAI analysis with quantified Activation Energy Ratio (AER), and TFLite INT8 deployment benchmarking within a single evaluation pipeline. Prior studies such as Barman et al. [5] and Ferentinos [6] evaluated CNN architectures for plant disease classification but did not incorporate statistical significance testing across multiple initializations or quantitative XAI validation, limiting the reproducibility and interpretability of their findings. Furthermore, the risk of evaluating models using only a single random seed initialization should not be overlooked. The standard deviations ranging from 0.005 to 0.010 observed in this study indicate that results from a single trial could deviate from the true mean by up to $\pm 1.0\%$. Therefore, the methodology presented here—comprising five repeated trials, results reported as mean $\pm$ SD, and Wilcoxon Signed-Rank testing—should be considered a minimum standard for such research, consistent with the recommendations of Picon et al. [34].

Regarding model interpretability, analysis using Grad-CAM++ [14] confirmed correct localization in 5 out of 6 disease types, with no instances of incorrect localization. This finding is consistent with Pal et al. [15] and confirms the proposal by Selvaraju et al. [13] that XAI is a vital component in building trust between users and AI systems. It also aligns with the work of Pei et al. [16], demonstrating that compact models can provide clear and reliable decision explanations even in resource-constrained systems. This transparency is particularly important in agricultural contexts, where farmer confidence in AI-driven recommendations directly influences adoption rates.

Practically, these findings offer significant opportunities for farmers in developing countries who often face resource and infrastructure constraints. MobileNetV3-Large, EfficientNetV2-B0, and NASNet-Mobile, with model sizes ranging from only 17–29 MB, can operate on standard smartphones without requiring internet connectivity or cloud

servers. Among these, EfficientNetV2-B0 stood out for its stability, exhibiting the lowest standard deviation (± 0.0054), consistent with the design principles of Tan and Le [10], which aim for high consistency under diverse data conditions. While NASNet-Mobile performed similarly to MobileNetV3-Large, its complex architectural cell structure may pose limitations during the TFLite conversion process, unlike MobileNetV3-Large, which Howard et al. [8] designed with mobile compatibility in mind from the outset.

Regarding economic impact, the platform's potential to detect Flower rusty spot and anthracnose 2–3 days earlier is vital for the Thai orchid export industry. Timely detection allows farmers to make management decisions before disease spread causes widespread impact, consistent with findings by Guo et al. [35] that automated plant disease diagnosis systems can effectively reduce losses from outbreaks in smart agriculture contexts.

While this study focused exclusively on *Dendrobium Sonia* orchids cultivated under controlled conditions in two Thai provinces, the proposed two-phase transfer learning pipeline is inherently architecture-agnostic and dataset-independent, suggesting strong potential for adaptation to other economically important crops such as roses, marigolds, and cassava. However, generalization to different climatic zones—particularly high-humidity tropical environments versus semi-arid conditions—would require domain adaptation strategies, including targeted data augmentation simulating varied lighting and weather conditions [24,25], or fine-tuning on locally collected datasets [34]. Such strategies would help mitigate potential distribution shifts between training and deployment environments [4], which prior work has identified as a key limitation of models trained under controlled conditions. Future validation across geographically diverse farms and plant species is therefore recommended to confirm the transferability of the proposed framework beyond the current experimental scope.

5. Conclusions

This research presents a comparative study of four CNN architectures—ResNet-50, MobileNetV3-Large, EfficientNetV2-B0, and NASNet-Mobile—for the classification of six types of *Dendrobium Sonia* orchid diseases under identical experimental conditions, combining statistical validation, XAI-based interpretability, and mobile deployment assessment within a unified framework.

Quantitatively, ResNet-50 achieved the highest overall accuracy (0.9896 ± 0.0097) and macro-F1-score (0.9894 ± 0.0098). Critically, the Wilcoxon Signed-Rank Test analysis confirmed that performance differences among all four architectures were statistically insignificant ($p > 0.05$), establishing that lightweight models—MobileNetV3-Large, EfficientNetV2-B0, and NASNet-Mobile—deliver classification performance statistically comparable to ResNet-50 while maintaining model sizes of only 17–29 MB and CPU inference latencies below 30 ms per image. Notably, the ResNet-50 vs. NASNet-Mobile comparison yielded a borderline p -value of 0.0625, which should be interpreted with caution given the limited number of seeds ($n = 5$).

Qualitatively, Grad-CAM++ analysis confirmed that activation energy was predominantly concentrated on pathological lesion regions across all architectures, with mean Activation Energy Ratio (AER) values ranging from 0.686 to 0.768—substantially higher than the 0.50 random-baseline threshold—providing empirical evidence that model decisions are grounded in diagnostically relevant features rather than background artifacts.

In terms of scientific novelty, this study advances reproducibility standards in plant disease classification research by systematically integrating five-seed statistical validation, Wilcoxon Signed-Rank significance testing, quantified Grad-CAM++ XAI analysis with AER metrics, and TFLite INT8 deployment benchmarking within a unified comparative

framework—directly addressing the single-run reproducibility limitations identified in prior works.

Based on these findings, MobileNetV3-Large was selected for Android-based mobile application development, offering an optimal balance between classification accuracy, model compactness, and real-time inference capability, directly addressing the practical needs of smallholder orchid farmers without access to laboratory infrastructure. Among the lightweight architectures evaluated, EfficientNetV2-B0 further distinguished itself through exceptional training stability ($SD \pm 0.0054$), making it a compelling alternative where consistency across diverse data conditions is prioritized.

Nevertheless, this research has certain limitations. The total dataset of 1832 images collected from ten farms across two provinces under controlled conditions may limit generalizability to diverse farm environments, natural lighting variations, and orchid varieties beyond *Dendrobium Sonia*. Future work should expand the dataset to at least 500 images per class under naturalistic conditions, investigate hybrid CNN-ViT architectures, apply knowledge distillation using ResNet-50 as the teacher model, explore domain adaptation strategies for deployment across diverse climatic zones, and validate models with 5-Fold Stratified Cross-Validation to strengthen the robustness and transferability of the proposed diagnostic framework toward practical deployment in the era of sustainable smart agriculture.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial Intelligence
AER	Activation Energy Ratio
XAI	Explainable Artificial Intelligence
CNN	Convolutional Neural Network
Grad-CAM++	Gradient-weighted Class Activation Mapping Plus Plus
IR	Imbalance Ratio
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
F1	F1-score
SD	Standard Deviation
NAS	Neural Architecture Search
ViT	Vision Transformer
TFLite	TensorFlow Lite
INT8	8-bit Integer (Quantized Representation)
MB	Megabyte
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative

References

1. Pianzo, M. *Country Profile Thailand*; International Association of Horticultural Producers (AIPH): Oxfordshire, UK, 2024. Available online: <https://aiph.org/wp-content/uploads/2025/02/Thailand-Report-FINAL-reduced.pdf> (accessed on 12 April 2026).
2. Wannakraijoj, S.; Loha-anu, M.; Wasuwat, K.; Wiraphan, R.; Rakpaiboolsombat, S.; Chaiwongrung, N.; Chaisuriyakul, Y.; Nawacharoen, W.; Triasri, N.; Thatkhon, T.; et al. *Orchid: Science and Art for Development*; Kasetsart University: Bangkok, Thailand, 2023. Available online: <https://kukr.lib.ku.ac.th/covid19/Detail/info/20017390> (accessed on 12 April 2026).
3. Saem Si Muang, K.; Saengpradit, K.; Pintong, A. Development of a Remote-Controlled Orchid Disease Inspection System with Image Processing Techniques for Precise Chemical Application in Standard Greenhouses. *Thai Soc. Agric. Eng. J.* **2016**, *22*, 7–20. Available online: <https://li01.tci-thaijo.org/index.php/TSAEJ/article/view/68886> (accessed on 23 March 2026).
4. Mohanty, S.P.; Hughes, D.P.; Salathé, M. Using deep learning for image-based plant disease detection. *Front. Plant Sci.* **2016**, *7*, 1419. [CrossRef] [PubMed]
5. Barman, U.; Choudhury, R.D.; Sahu, D.; Barman, G.G. Comparison of convolution neural networks for smartphone image-based real-time classification of citrus leaf disease. *Comput. Electron. Agric.* **2020**, *177*, 105661. [CrossRef]
6. Ferentinos, K.P. Deep learning models for plant disease detection and diagnosis. *Comput. Electron. Agric.* **2018**, *145*, 311–318. [CrossRef]
7. Howard, A.G.; Zhu, M.; Chen, B.; Kalenichenko, D.; Wang, W.; Weyand, T.; Andreetto, M.; Adam, H. MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications. *arXiv* **2017**, arXiv:1704.04861. [CrossRef]
8. Howard, A.; Sandler, M.; Chu, G.; Chen, L.-C.; Chen, B.; Tan, M.; Wang, W.; Zhu, Y.; Pang, R.; Vasudevan, V.; et al. Searching for MobileNetV3. In *Proceedings of the IEEE/CVF International Conference on Computer Vision, Seoul, South Korea, 27 October–2 November 2019*; IEEE: Piscataway, NJ, USA, 2019; pp. 1314–1324. [CrossRef]
9. Sandler, M.; Howard, A.; Zhu, M.; Zhmoginov, A.; Chen, L.-C. MobileNetV2: Inverted Residuals and Linear Bottlenecks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, Salt Lake City, UT, USA, 18–23 June 2018*; IEEE: Piscataway, NJ, USA, 2018; pp. 4510–4520. [CrossRef]
10. Tan, M.; Le, Q.V. EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. In *Proceedings of the 36th International Conference on Machine Learning, PMLR, Long Beach, CA, USA, 9–15 June 2019*; Volume 97, pp. 6105–6114. Available online: <https://proceedings.mlr.press/v97/tan19a.html>.
11. Tan, M.; Le, Q.V. EfficientNetV2: Smaller Models and Faster Training. In *Proceedings of the 38th International Conference on Machine Learning, PMLR, Virtual, 18–24 July 2021*; Volume 139, pp. 10096–10106. Available online: <https://proceedings.mlr.press/v139/tan21a.html>.
12. Zoph, B.; Vasudevan, V.; Shlens, J.; Le, Q.V. Learning transferable architectures for scalable image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Salt Lake City, UT, USA, 18–23 June 2018*; IEEE: Piscataway, NJ, USA, 2018; pp. 8697–8710. [CrossRef]

13. Selvaraju, R.R.; Cogswell, M.; Das, A.; Vedantam, R.; Parikh, D.; Batra, D. Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE International Conference on Computer Vision, Venice, Italy, 22–29 October 2017*; IEEE: Piscataway, NJ, USA, 2017; pp. 618–626. [\[CrossRef\]](#)
14. Chattopadhyay, A.; Sarkar, A.; Howlader, P.; Balasubramanian, V.N. Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks. In *Proceedings of the IEEE Winter Conference on Applications of Computer Vision, Lake Tahoe, NV, USA, 12–15 March 2018*; IEEE: Piscataway, NJ, USA, 2018; pp. 839–847. [\[CrossRef\]](#)
15. Pal, C.; Islam Karmakar, S.; Mukherjee, I.; Chakrabarti, P.P. A lightweight and explainable CNN model for empowering plant disease diagnosis. *Sci. Rep.* **2025**, *15*, 30720. [\[CrossRef\]](#)
16. Pei, Z.; Meng, Z.; Diao, T.; Miao, P.; Meng, Y.; Tan, C. A visually explainable dynamic similarity network for few-shot classification. *J. Artif. Intell. Soft Comput. Res.* **2026**, *16*, 237–256. [\[CrossRef\]](#)
17. Islam, M.N.; Hossain, E.; Hossain, M.; Andersson, K. Plant Disease Detection and Classification by Deep Learning. *Plants* **2019**, *8*, 468. [\[CrossRef\]](#) [\[PubMed\]](#)
18. Correa-Tome, F.E.; Sanchez-Yanez, R.E.; Ayala-Ramirez, V. Automatic Optimization of Deep Learning Training Through Feature-Aware-Based Dataset Splitting. *Algorithms* **2024**, *17*, 106. [\[CrossRef\]](#)
19. Kantipudi, M.V.V.P.; Sivasankari, G.; Opala, C.; Rao Sai, K. Stratified Sampling-Based Deep Learning Approach to Increase Prediction Accuracy of Unbalanced Dataset. *Electronics* **2023**, *12*, 4423. [\[CrossRef\]](#)
20. Hossain, E.; Islam, M.K.; Hossain, M.; Andersson, K.; Hossain, M.S. Exploring the Interplay of Dataset Size and Imbalance on CNN Performance in Healthcare: Using X-rays to Identify COVID-19 Patients. *Diagnostics* **2024**, *14*, 1727. [\[CrossRef\]](#) [\[PubMed\]](#)
21. Tang, Z.; Xie, Q.; Zhang, Y.; Zhou, H. Review on Convolutional Neural Network (CNN) Applied to Plant Leaf Disease Classification. *Agriculture* **2021**, *11*, 707. [\[CrossRef\]](#)
22. Tugrul, B.; Elfatimi, E.; Eryigit, R. Convolutional Neural Networks in Detection of Plant Leaf Diseases: A Review. *Agriculture* **2022**, *12*, 1192. [\[CrossRef\]](#)
23. Waheed, A.; Goyal, M.; Gupta, D.; Khanna, A.; Al-Turjman, F.; Pinheiro, P.R. Plant Disease Classification: A Comparative Evaluation of Convolutional Neural Networks and Deep Learning Optimizers. *Plants* **2020**, *9*, 1319. [\[CrossRef\]](#) [\[PubMed\]](#)
24. Min, B.; Kim, T.; Shin, D.; Shin, D. Data Augmentation Method for Plant Leaf Disease Recognition. *Appl. Sci.* **2023**, *13*, 1465. [\[CrossRef\]](#)
25. Lee, S.; Lee, S. Efficient Data Augmentation Methods for Crop Disease Recognition in Sustainable Environmental Systems. *Big Data Cogn. Comput.* **2025**, *9*, 8. [\[CrossRef\]](#)
26. Chen, X.; Wang, L.; Chattopadhyay, P. Vision Transformers for Image Classification: A Comparative Survey. *Technologies* **2025**, *13*, 32. [\[CrossRef\]](#)
27. Dosovitskiy, A.; Beyer, L.; Kolesnikov, A.; Weissenborn, D.; Zhai, X.; Unterthiner, T.; Dehghani, M.; Minderer, M.; Heigold, G.; Gelly, S.; et al. An Image is Worth 16 × 16 Words: Transformers for Image Recognition at Scale. *arXiv* **2021**. [\[CrossRef\]](#)
28. da Silva, F.M.; Tavares, J.M.R.S.; Lopes, A.M.; Silva, A.R. Cross-dataset benchmarking of deep learning models for surface defect classification in metal parts. *Appl. Sci.* **2026**, *16*, 3022. [\[CrossRef\]](#)
29. Zhou, B.; Khosla, A.; Lapedriza, A.; Oliva, A.; Torralba, A. Learning deep features for discriminative localization. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Las Vegas, NV, USA, 27–30 June 2016*; IEEE: Piscataway, NJ, USA, 2016; pp. 2921–2929. [\[CrossRef\]](#)
30. Otsu, N. A threshold selection method from gray-level histograms. *IEEE Trans. Syst. Man Cybern.* **1979**, *9*, 62–66. [\[CrossRef\]](#)
31. Picard, D. Torch.manual_seed(3407) is all you need: On the Influence of Random Seeds in Deep Learning Architectures for Computer Vision. *arXiv* **2021**, arXiv:2109.08203. [\[CrossRef\]](#)
32. Henderson, P.; Islam, R.; Bachman, P.; Pineau, J.; Precup, D.; Meger, D. Deep reinforcement learning that matters. In *Proceedings of the AAAI Conference on Artificial Intelligence, New Orleans, LA, USA, 2–7 February 2018*; AAAI Press: Washington, DC, USA, 2018; pp. 3207–3214. [\[CrossRef\]](#)
33. Zhang, S.; Huang, W.; Zhang, C. Plant Disease Detection Using Deep Learning: A Review. *Artif. Intell. Agric.* **2019**, *3*, 33–41. [\[CrossRef\]](#)
34. Picon, A.; Alvarez-Gila, A.; Seitz, M.; Ortiz-Barredo, A.; Echazarra, J.; Johannes, A. Deep convolutional neural networks for mobile capture device-based crop disease classification in the wild. *Comput. Electron. Agric.* **2019**, *161*, 280–290. [\[CrossRef\]](#)
35. Guo, Y.; Zhang, J.; Yin, C.; Hu, X.; Zou, Y.; Xue, Z.; Wang, W. Plant disease identification based on deep learning algorithm in smart farming. *Discret. Dyn. Nat. Soc.* **2020**, *2020*, 2479172. [\[CrossRef\]](#)

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